# Full Stack Web Development - Capstone

Presentation

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# Slide 1: Title Slide

- Content:
  - Title: Library Management System (LMS) - Capstone Project
  - Subtitle: Full Stack Web Development Project
  - Presented by: [Learner's Name]
  - Date: [Presentation Date]
- Presentation Hints:
  - Briefly introduce yourself and the project.
  - Mention that the LMS is a web application designed to streamline library operations and enhance user experience.

# Slide 2: Project Overview

- Mention About:
  - Web application for managing library operations.
  - Two main user roles: Librarian (Admin) and Member.
  - Built using Spring Boot (Backend), React (Frontend), and MySQL (Database).
- Presentation Hints:
  - Explain the purpose of the LMS.
  - Highlight the two user roles and their functionalities.
  - Mention the technologies used.

# Slide 3: Project Goals

- Mention about:
  - Develop a fully functional web application.
  - Enable librarians to manage books and members efficiently.
  - Provide a user-friendly interface for members to search for books and track borrowing history.
  - Create a robust and scalable system.
- Presentation Hints:
  - Emphasize the importance of creating a functional and scalable system.
  - Mention the focus on user experience for both librarians and members.

# Slide 4: Key Features

- Mention about:
  - User Management: Registration, login, profile management, password reset.
  - Member Management: Register, edit, delete, and search members.
  - Book Management: Add, update, delete, and search books.
  - Lending Management: Borrow, return, reserve books, and track due dates.
- Presentation Hints:
  - Briefly explain each feature.
  - Highlight how these features will improve library operations.

# Slide 5: Business Rules

- Mention some rules if you have implemented:
  - Membership Rules: Max 3 books, 1-year validity, valid membership required.
  - Lending Rules: 14-day loan, max 2 renewals, no overdue books or fines > $10.
  - Fine Calculation: $0.50 per day, max $20 per book.
- Presentation Hints:
  - Explain the rules governing memberships and lending.
  - Mention how fines are calculated.

# Slide 6: Technical Requirements

- Mention about:
  - **Backend**: Spring Boot, RESTful API, MySQL, Spring Security, JPA/Hibernate.
  - **Frontend**: React.js, React Router, React Hooks, Bootstrap/Material-UI, Axios.
  - **Database**: Proper relationship mapping, referential integrity, indexing, data validation.
- Presentation Hints:
  - Highlight the technologies used for backend, frontend, and database.
  - Mention the importance of security and data integrity.

# Slide 7: System Architecture

- Mention about
  - Architecture type you chose: ex MVC

- Presentation Hints:
  - Describe how MVC helped you in system implementation

# Slide 8: Screenshots of key features

- Admin Portal
  - User Management
  - Book Management
- Member Portal
  - Book Search
  - Borrow History
- Presentation Hints:
  - Describe role of each portal
  - Talk on various features supported under each portal

# Slide 9: Database & System Designs

- Add screenshots of ERDs and explain relationships
- Wireframe/prototype design of
  - Overall Library System
  - Features implemented like user registration, login etc
- Presentation Hints:
  - Describe main entities and relationships
  - With wireframes/prototype explain the flow of the system

# Slide 10: API endpoints

- Mention about
  - The endpoints used by admin and members
  - Endpoints for actions that user and admin can perform (CRUD operations)

# Slide 11: Test Cases - UATs

- Mention about
  - Mention various tests you have considered. For example, Functional UATs for user management, book management etc
  - Screenshot(s) of output from various test runs
- Presentation Hints:
  - Talk about how tests are carried out for user registration, login etc

# Slide 12: Challenges and Solutions

- Mention about:
  - Integrating frontend and backend.
  - Implementing role-based access control.
  - Handling fine calculation and overdue books.
  - Ensuring data validation and security.
- Presentation Hints:
  - Discuss the challenges encountered during the project.
  - Explain how these challenges were overcome.

# Slide 13: Future Enhancements

- Mention about:
  - Mobile app integration.
  - Advanced search and recommendation system.
  - Integration with external libraries or APIs.
  - Enhanced reporting and analytics for librarians.
- Presentation Hints:
  - Discuss potential future improvements to the LMS.
  - Mention how these enhancements could further improve the system.

# Slide 14: Conclusion

- Project Summary
- Key achievements
- Q&A

# Thank You!

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